Year 7 Mathematics

Unit 4: Fractions

4.2 + 4.5 Adding and Subtracting Fractions

Adding and Subtracting Fractions

STARTER · 5 MIN

Date: _____ **Title: Adding and Subtracting Fractions**

LEARNING OBJECTIVES

- To be able to add and subtract fractions

STARTER *Answer in your book. No calculators.*

1. Write $\frac{6}{10}$ in simplest form

2. Write $\frac{12}{14}$ in simplest form

3. Find the LCM of 6 and 8

4. Find the LCM of 5 and 10

Starter — Answers

STARTER

Tick or correct in green pen

1 $\frac{3}{5}$

2 $\frac{6}{7}$

3 24

4 10

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can write an improper fraction as a mixed number and add or subtract with the same denominator	☐
SILVER	I can use a common denominator to add and subtract fractions	☐
GOLD	I can add and subtract mixed numbers and simplify the answer	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Adding and Subtracting Fractions

INTRODUCTION · 3 MIN

Fractions can be added or subtracted only when the denominators are the same.

Improper fraction

The top is bigger than the bottom.

$$\frac{7}{4} = \frac{4}{4} + \frac{3}{4} = 1\ \frac{3}{4}$$

Same denominator

Add or subtract the tops only.

$$\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$$

Common denominator

Change to the same bottom (LCM).

$$\frac{2}{5} = \frac{4}{10}$$

Mixed numbers

Wholes and fractions together.

$$2\ \frac{3}{4} = \frac{11}{4}$$

Examples 1 and 2 — Same Denominator and Mixed Numbers

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Same bottom: add the tops 12 ÷ 5 = 2 r 2

Example 1

Work out

$$\frac{2}{9} + \frac{1}{9}$$

Give it in simplest form.

Example 2

Write as a mixed number

$$\frac{12}{5}$$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Add the numerators, keep the bottom

$$\frac{2}{9} + \frac{1}{9} = \frac{3}{9}$$

$$= \mathbf{\frac{1}{3}}$$

Tip: never add the denominators.

Example 2

How many wholes? What is left?

$12 \div 5 = 2$ remainder 2

$$\mathbf{\frac{12}{5} = 2\ \frac{2}{5}}$$

Tip: 5 fifths make one whole.

Task 1

PRACTICE · 9 MIN

BRONZE

Work out. Give answers in simplest form

Hint Same bottom: add the tops 12 ÷ 5 = 2 r 2

1 $\frac{1}{3} + \frac{1}{3}$

2 $\frac{3}{7} + \frac{3}{7}$

3 $1 - \frac{3}{5}$

4 $\frac{7}{12} - \frac{5}{12}$

5 Write as a mixed number

$\frac{31}{7}$

★ Write as a mixed number

$\frac{23}{3}$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $\frac{2}{3}$

2 $\frac{6}{7}$

3 $\frac{2}{5}$

4 $\frac{2}{12} = \frac{1}{6}$

5 $4\frac{3}{7}$

★ $7\frac{2}{3}$

Examples 3 and 4 — Finding a Common Denominator

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint
Find the LCM of the bottoms
Same change top and bottom

Example 3

Work out

$$\frac{3}{10} + \frac{2}{5}$$

Example 4

Work out

$$\frac{5}{6} + \frac{2}{3}$$

Give a mixed number.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Change fifths into tenths

$$\frac{2}{5} = \frac{4}{10}$$

$$\mathbf{\frac{3}{10} + \frac{4}{10} = \frac{7}{10}}$$

Tip: 5 × 2 = 10, so top × 2 as well.

Example 4

Sixths, then a mixed number

$$\frac{2}{3} = \frac{4}{6}$$

$$\frac{5}{6} + \frac{4}{6} = \frac{9}{6}$$

$$\mathbf{= 1\ \frac{1}{2}}$$

Tip: 9 sixths = 1 whole and 3 sixths.

Task 2

PRACTICE · 9 MIN

SILVER

Work out. Give answers in simplest form

Hint Find the LCM of the bottoms
Same change top and bottom

1 $\frac{4}{9} + \frac{1}{3}$

2 $\frac{7}{10} - \frac{1}{5}$

3 $\frac{11}{12} - \frac{3}{4}$

4 $\frac{1}{2} + \frac{1}{3}$

5 $\frac{3}{4} + \frac{11}{16}$

★ $\frac{1}{3} + \frac{8}{9}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $\frac{7}{9}$

2 $\frac{1}{2}$

3 $\frac{1}{6}$

4 $\frac{5}{6}$

5 $1\frac{7}{16}$

★ $1\frac{2}{9}$

Examples 5 and 6 — Adding and Subtracting Mixed Numbers

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Wholes, then fractions Or use improper fractions

Example 5

(2 marks)

Work out

$$2\ \frac{3}{4} + 1\ \frac{1}{8}$$

Example 6

(3 marks)

Work out

$$5\ \frac{1}{4} - 2\ \frac{3}{8}$$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Add wholes, then fractions

Wholes: $2 + 1 = 3$

$$\frac{6}{8} + \frac{1}{8} = \frac{7}{8}$$

Answer $= 3\ \frac{7}{8}$

Tip: make the denominators match first.

Example 6

Change to improper fractions

$$5\ \frac{1}{4} = \frac{21}{4} = \frac{42}{8}$$

$$\frac{42}{8} - \frac{19}{8} = \frac{23}{8}$$

$$= 2\ \frac{7}{8}$$

Tip: $2 \times 8 + 3 = 19$ eighths.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: give answers in simplest form

Hint Wholes, then fractions
Or use improper fractions

1 $1\frac{2}{3} + 3\frac{5}{6}$

2 $3\frac{3}{6} + 2\frac{2}{3}$

3 $4\frac{2}{3} - 2\frac{1}{6}$

4 $5\frac{1}{4} - 3\frac{5}{8}$

5 $3\frac{7}{10} + 2\frac{3}{5}$

★ $3\frac{1}{3} - 1\frac{3}{6}$

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $5\frac{1}{2}$

2 $6\frac{1}{6}$

3 $2\frac{1}{2}$

4 $1\frac{5}{8}$

5 $6\frac{3}{10}$

★ $1\frac{5}{6}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write $\frac{11}{4}$ as a mixed number.

SILVER Work out $\frac{2}{7} + \frac{3}{7}$.

GOLD Work out $\frac{5}{6} - \frac{1}{3}$.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $2\frac{3}{4}$

SILVER $\frac{5}{7}$

GOLD $\frac{5}{6} - \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$

Exam Practice — Same Denominator

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Same bottom: add the tops 12 ÷ 5 = 2 r 2

Question 1

(2 marks)

Give each answer in simplest form.

(a) $\frac{3}{8} + \frac{1}{8}$ (1)

.....

(b) $1 - \frac{5}{9}$ (1)

.....

Question 2

(2 marks)

Write as mixed numbers.

(a) $\frac{17}{5}$ (1)

.....

(b) $\frac{11}{4}$ (1)

.....

Exam Practice — Different Denominators

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint
Find the LCM of the bottoms
Same change top and bottom

Question 3

(2 marks)

Work out

$$\frac{2}{3} + \frac{1}{6}$$

Give your answer in its simplest form.

.....

Question 4

(2 marks)

Work out

$$\frac{3}{4} - \frac{2}{5}$$

.....

Exam Practice — Mixed Numbers

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Wholes, then fractions
Or use improper fractions

Question 5

(2 marks)

Work out

$1\frac{2}{3} + 2\frac{1}{2}$

.....

Question 6

(3 marks)

A plank is $4\frac{1}{2}$ m long.

A piece $1\frac{3}{4}$ m long is cut off.

How long is the plank now?

..... m

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $\frac{4}{8} = \frac{1}{2}$ — B1

(b) $\frac{4}{9}$ — B1

Tip: 1 = 9 ninths.

Question 2

(a) $3\frac{2}{5}$ — B1

(b) $2\frac{3}{4}$ — B1

Tip: divide, the remainder goes on top.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$\frac{2}{3} = \frac{4}{6}$ **M1**

$\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$ **A1**

Tip: 6 is a multiple of 3.

Question 4

$\frac{15}{20} - \frac{8}{20} = \frac{7}{20}$ **M1 A1**

Tip: the LCM of 4 and 5 is 20.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$3 + \frac{4}{6} + \frac{3}{6} = 3 + \frac{7}{6}$ M1

$= 4\ \frac{1}{6}$ A1

Tip: add wholes, then fractions.

Question 6

$\frac{18}{4} - \frac{7}{4} = \frac{11}{4}$ M1 M1

$= 2\ \frac{3}{4}$ m A1

Tip: improper fractions avoid borrowing.